PHYS101/Fall 2021-2022 Midterm I

Physics 101 No: _____

Name: _____

(On my honor I have neither given nor received any unauthorized assistance on this exam.)

- This exam is out of 100 points.
- This exam is for 100 minutes and has five problems (Check this now carefully.)
- This is a closed-book/notes/calculator/laptop exam.
- Read the questions carefully and write down your solutions in the space provided neatly and clearly. Write down your final results in the boxes at the end of each question.
- Answers without work shown or proper explanation will not be given any credit.

(Leave blank)

P1	P2	P3	P4	P5	Total

Formula Sheet

Scalar Product:

$c = \vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$

$c = AB \cos\phi$

Vector Product:

$\vec{C} = \vec{A} \times \vec{B}$

$C = AB \sin\phi$

$C_x = A_y B_z - A_z B_y$

$C_y = A_z B_x - A_x B_z$

$C_z = A_x B_y - A_y B_x$

Straight-line motion:

$$v_{av-x} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{\boldsymbol{\Delta x}}{\boldsymbol{\Delta t}}$$

$$v_x = \lim_{\Delta t \to 0} \frac{\boldsymbol{\Delta x}}{\boldsymbol{\Delta t}} = \frac{dx}{dt}$$

$$a_{av-x} = \frac{v_{2x} - v_{1x}}{t_2 - t_1} = \frac{\boldsymbol{\Delta v_x}}{\boldsymbol{\Delta t}}$$

$$a_x = \lim_{\Delta t \to 0} \frac{\boldsymbol{\Delta v_x}}{\boldsymbol{\Delta t}} = \frac{dv_x}{dt}$$

Constant x-acceleration:

$v_x = v_{0x} + a_x t$

$x = x_0 + v_{0x} t + \frac{1}{2} a_x t^2$

$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$

$x - x_0 = \left(\frac{v_{0x} + v_x}{2}\right) t$

Motion in three dimensions:

$\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$

$$\vec{v}_{av} = \frac{\vec{r}_2 - \vec{r}_1}{t_2 - t_1} = \frac{\boldsymbol{\Delta \vec{r}}}{\boldsymbol{\Delta t}}$$

$$\vec{v} = \lim_{\Delta t \to 0} \frac{\boldsymbol{\Delta \vec{r}}}{\boldsymbol{\Delta t}} = \frac{d\vec{r}}{dt}$$

$$\vec{a}_{av} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1} = \frac{\boldsymbol{\Delta \vec{v}}}{\boldsymbol{\Delta t}}$$

$$\vec{a} = \lim_{\Delta t \to 0} \frac{\boldsymbol{\Delta \vec{v}}}{\boldsymbol{\Delta t}} = \frac{d\vec{v}}{dt}$$

Circular motion:

$$a_{\text{rad}} = \frac{v^2}{R} \quad a_{\text{rad}} = \frac{4\pi^2 R}{T^2}$$

$\sin 36.9^\circ = \cos 53.1^\circ = 0.6$

$\sin 53.1^\circ = \cos 36.9^\circ = 0.8$

Problem (1) (20 points)

The position of an object with mass $m = 2$kg is given as a function of time: $\vec{r}(t) = (t^2-2)\hat{\imath}+(3t-1)\hat{\jmath}$, where r is in meters.

a.) (5 pts.) Find the velocity of the, object at $t = 2$ seconds.

b.) (5 pts.) Find the acceleration of the, object at $t = 2$ seconds.

c.) (5 pts.) Find the angle θ between the velocity and the acceleration at $t = 2$ seconds. (Find $\sin\theta$ or $\cos\theta$.)

d.) (5 pts.) Find the kinetic energy of the ,object at $t = 2$ seconds. (Express the vector quantities in terms of unit vectors.)

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a-) $\vec{v}(t) = \dfrac{d\vec{r}(t)}{dt} = 2t\,\hat{\imath} + 3\,\hat{\jmath} \Rightarrow \vec{v}(2) = (4\hat{\imath} + 3\hat{\jmath})$ m/s

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b-) $\vec{a}(t) = \dfrac{d\vec{v}(t)}{dt} = 2\hat{\imath} \Rightarrow \vec{a}(2) = (2\hat{\imath})$ m/s^2

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c-) $\vec{v}(2)\cdot\vec{a}(2) = |\vec{v}(2)|\,|\vec{a}(2)|\cos\theta = (5)(2)\cos\theta$

$\vec{v}(2)\cdot\vec{a}(2) = 8$ $\boxed{\cos\theta = 4/5}$

d-) $K(t) = \frac{1}{2} m v^2(t) \Rightarrow K(2) = \frac{1}{2}(2)(5)^2 = 25$ Joule

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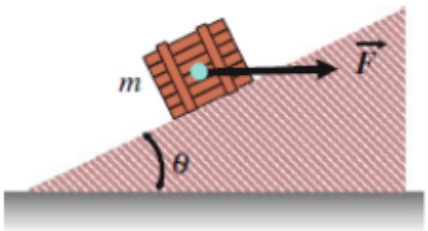
Problem (2) (20 points)



A block of mass m moves *up* with *constant speed* on an inclined plane. A constant horizontal force F acts on the box as shown in the figure.

a.) (5 pts.) Draw a free body diagram for m.

b.) (15 pts.) Derive an expression for the coefficient of kinetic friction between the block and the surface, in terms of F, m, g, and θ.

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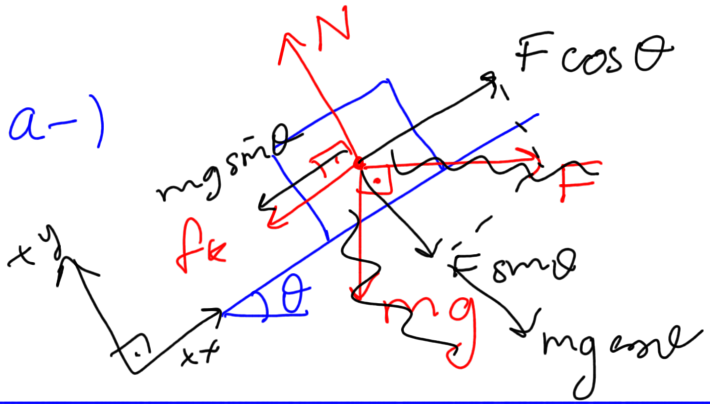
a-)



$f_k = \mu_k N$

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b-) $\sum F_y = ma_y \Rightarrow N - F\sin\theta - mg\cos\theta = 0$

$\boxed{a_y = 0}$

$N = F\sin\theta + mg\cos\theta$

$\sum F_x = ma_x \Rightarrow F\cos\theta - mg\sin\theta - \mu_k N = 0$

$\boxed{a_x = 0 \text{ constant speed!}}$

$$\boxed{\mu_k = \frac{F\cos\theta - mg\sin\theta}{F\sin\theta + mg\cos\theta}}$$

Problem (3) (25 points)

A constant force $\vec{F} = 4\hat{\imath} - 10\hat{\jmath}$, in Newton, acts on an object with mass $m = 2$kg. At $t = 0$ the object is at the origin ($x = 0, y = 0$), and has an initial velocity: $\vec{v}_0 = 4\hat{\imath} + 10\hat{\jmath}$, in m/s.

a.) (5 pts.) When does the object reach maximum height?

b.) (5 pts.) Find the maximum height.

c.) (5 pts.) Find the velocity vector at maximum height.

d.) (5 pts.) Find the position vector at maximum height.

e.) (5 pts.) Find the work done on the object from the initial position to the maximum height.

a-) $a_x(t) = \frac{F_x(t)}{m} = 2\ m/s^2 \qquad a_y(t) = \frac{F_y(t)}{m} = -5\ m/s^2$

$v_y(t) = v_y(0) + \int_0^t a_y(t)\,dt = 10 + \int_0^t -5\,dt = 10 - 5t$

at maximum height: $v_y(t_{max}) = 0 \Rightarrow 10 - 5t_{max} = 0$

$t_{max} = 2$ sec

b-) $y(t) = y(0) + \int_0^t v_y(t)\,dt \Rightarrow 0 + \int_0^t (10 - 5t)\,dt = 10t - \frac{5}{2}t^2$

$y_{max} = y(t_{max}) = y(2) = (10)(2) - \frac{5}{2}(2)^2 = 10\,m$

c-) $v_x(t) = v_x(0) + \int_0^t a_x(t)\,dt = 4 + \int_0^t 2\,dt = 4 + 2t$

$\vec{v}(t) = v_x(t)\,\hat{\imath} + v_y(t)\,\hat{\jmath} \Rightarrow$ for $t_{max} = 2 \quad \vec{v}(2) = (8\,\hat{\imath})\ m/s$

d-) $x(t) = x(0) + \int_0^t v_x(t)\,dt = 0 + \int_0^t (4 + 2t)\,dt = 4t + t^2$

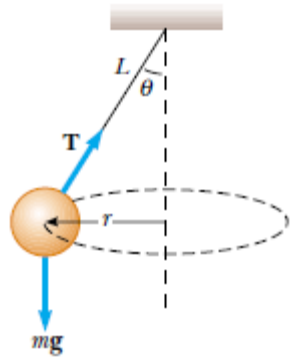
$\vec{r}(t) = x(t)\,\hat{\imath} + y(t)\,\hat{\jmath} \Rightarrow$ for $t_{max} = 2 \quad \vec{r}(2) = (12\,\hat{\imath} + 10\,\hat{\jmath})\ m$

e-) $W_{TOT} = \Delta K = K_f - K_i$

$\boxed{W_{TOT} = -52 \text{ Joule}}$

$K_f = \frac{1}{2}(2)(64) \qquad K_i = \frac{1}{2}(2)(116)$

Problem (4) (20 points)



A conical pendulum consists of a small object of mass m which is suspended from a string of length L. The object rotates in a horizontal circle at constant speed V. T is the tension on the string, and g is the gravitational acceleration.

a.) (5 pts.) Draw a free body diagram for m.

b.) (15 pts.) Derive an expression for the speed of the object V, in terms of L, g, and θ.

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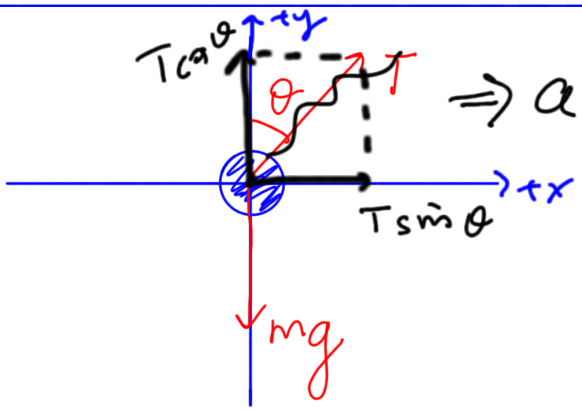
a-)



$$\Rightarrow a_{Radial} = \frac{V^2}{r}$$

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b-) $\Sigma F_y = m a_y \Rightarrow$ $\quad T\cos\theta - mg = 0$

$a_y = 0$

$\Sigma F_x = m a_x \Rightarrow$ $\quad T\sin\theta = m\frac{V^2}{r}$

$a_x = a_{Radial}$

$$T\sin\theta = m\frac{V^2}{r}$$

$$\div \quad T\cos\theta = mg$$

$$\tan\theta = \frac{V^2}{rg} \qquad r = L\sin\theta$$

$$V = \sqrt{gL\tan\theta\sin\theta}$$

Problem (5) (15s points)

An object moves in one dimension (in $+x$ direction) through a viscous medium, and experiences a resistive force that opposes its motion. The resistive force is proportional to the velocity of the object i.e.

$$\vec{F} = -bV\hat{\imath}$$

At $t = 0$ the particle has velocity $\vec{V} = V_0\hat{\imath}$.

a.) (3 pts.) What is the unit of the constant b in SI units kilograms, meters, seconds ?

b.) (8 pts.) Find the velocity of the object as a function of time.

c.) (4 pts.) Find the acceleration of the object as a function of time.

Hint: $\int \frac{du}{u} = \ln(u)$, $\int \frac{du}{u^2} = \frac{-1}{u}$

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a-) $F \to kg\, m/s^2 \qquad V \to m/s$

$[kg\, m/s^2] \equiv [b][m/s] \qquad [b] \equiv kg/s$ "unit for b"

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b-) $a(t) = \frac{F(t)}{m} = \frac{-bV(t)}{m}$

$\frac{dV(t)}{dt} = a(t) \Rightarrow \frac{dV(t)}{dt} = -\frac{b}{m}V(t)$

$\Rightarrow \frac{dV(t)}{V(t)} = -\frac{b}{m}dt$ take the integral of both sides

$\int \frac{dV(t)}{V(t)} = \int -\frac{b}{m}dt \Rightarrow \ln[V(t)] = -\frac{bt}{m} + C$

$\Rightarrow V(t) = e^{-\frac{bt}{m}} \cdot e^{C}$

we can call $e^{C} = A$

$V(t) = A e^{-bt/m}$ at $t=0$ $V(0) = V_0 \Rightarrow A = V_0$

$V(t) = V_0 e^{-bt/m}$

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c-) $a(t) = \frac{dV(t)}{dt} = -\frac{bV_0}{m} e^{-bt/m}$